

Table 3. Regression results.

Marketing Factor	Model I: Volume of sale model*	Model II: Value of sale model
Ln_Price	-0.005 (-0.527)	0.420 (71.380)***
Ln_Distribution points	0.001(0.022)	0.001 (-0.014)
Ln_Number of customers	1.002 (16.277)***	0.616 (16.367)***
Constant	-0.024 (-0.181)	-0.027 (-0.205)
Model diagnostics:	<i>Adjusted R²=0.978; F=13820 (df=101;sig@0.000)</i>	<i>Adjusted R²=0.988; F=37040 (df=101;sig@0.000)</i>

*T-values in parenthesis; *** denotes significance at 1% level.

Source: SPSS Regression output from dunnhumby data (2011-2013).

for exported Ghanaian yam on UK supermarket. It is worth noting that retail sales value driven by price increases is not sustainable unless it occurs with improved product quality. The obvious focus of the industry will be to attract more customers to the market.

Marketing strategies and export policy implications

In terms of marketing implications the findings of the study could inform what marketing strategies will be effective for future success. Managers of exported yam portfolio have the option of concentrating efforts on existing customers and/or exploring market expansion. Shopper loyalty indicators such as repeat rate of purchasing yam, amount of money spent per customer and number of tubers purchased per customer collectively support the option of focusing on improving existing customer satisfaction towards maintaining and enhancing customer loyalty. There is however the alternative to aggressively pursue strategies to attract new customers through in-store as well as external marketing communication methods, such as sales promotion, in-store tastings, provision of more nutritional information and introduction of a specialised isle for exotic roots and tuber products.

The findings of this study also have implications for yam export supply chain management. Specifically, the yam supply chain ought to be developed efficiently by taken into account exported yam shopper insight from the export market instead of operating the existing one which is based on assumptions about who buys Ghanaian yam in the UK. Stakeholders tasked with managing and promoting yam exports ought to explore existing avenues to access export market shopper insights based on actual purchasing behaviour such as dunnhumby loyalty card data used for the analysis in this paper. There is also a latent potential for growth as supermarkets in foreign export destinations such as UK continue to incorporate more ethnic foods in their product portfolio. However, such enabling market environment would require further quality improvements in production, packaging and delivery standards because of the ever rising levels of

competition and requirements from buyers in Europe and Northern America.

Limitations and areas for future research

The findings of the current study are based on a year-on-year comparison and regression analysis of two years of dunnhumby loyalty card data from Tesco, which is representative of the UK grocery food shopper. The data is however limited in the sense that the greater amount of Ghanaian yam exported are retailed by relatively smaller Afro-Caribbean and Asian shops across the UK. Thus, a study capturing yam customers from smaller ethnic shops would provide important insight that will complement the findings of this study. Secondly, the model for the regression analysis concentrated on number of customers, price and distribution; other marketing factors which are not accounted for in this study could contribute to supermarket retail sales growth of yam in the UK. Therefore, assessing the roles of additional factors such as promotion and provenance (country of produce of yam) could provide useful insights that will inform adopting appropriate marketing strategies to sustain the yam export market.

REFERENCES

- Aidoo R, Nimoh F, Bakang JA, Ohene-Yankyer K, Fialor SC, Mensah JO, Abaidoo RC (2012). Estimation of Margins and Efficiency in the Ghanaian Yam Marketing Chain. *Asian J. Agric. Rural Dev.*, 2(2): pp.226-234.
- Ajzen I (1991). The theory of planned behaviour. *Org. Beh. Human Decision Process*, 50(1): pp.179-211.
- Ajzen I, Fishbein M (1980). *Belief, Attitude, Intention, and Behaviour: An Introduction to Theory and Research*, Addison-Wesley, Reading.
- Amanze NJ, Agbo NJ, Eke-Okoro ON, Njoku DN (2011). Selection of yam seeds from open pollination for adoption in yam (*Dioscorea rotundata* Poir) production zones in Nigeria. *J. Plant Breeding and Crop Sci.* 3(4): pp.68-73.